

Project Initialization and Planning Phase

Date	6 July 2026
Team ID	SWTID-2026-8977
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution, EduGenie: Google Gemini Powered Learning Assistant, aims to transform the learning experience by leveraging Generative AI to provide intelligent educational assistance. The platform addresses the challenges students and self-learners face in understanding complex concepts, finding reliable learning resources, and preparing effectively for examinations. By integrating Google Gemini AI, EduGenie delivers accurate answers, simplified explanations, AI-generated quizzes, educational summaries, and personalized learning paths through an interactive web-based interface.

Project Overview	
Objective	The primary objective of EduGenie is to enhance learning by providing AI-powered educational assistance through Google Gemini. The system enables students to ask questions, understand concepts, generate quizzes, summarize educational content, and receive personalized learning recommendations quickly and accurately.
Scope	The project focuses on developing a web-based educational assistant using FastAPI, Google Gemini API, and a responsive HTML/CSS interface. It supports intelligent question answering, concept explanation, quiz generation, text summarization, and personalized learning paths for students across multiple academic subjects.
Problem Statement	

Description	Students often struggle to understand complex topics, search through multiple educational resources, and create effective study plans. Traditional learning methods are time-consuming and do not provide personalized assistance or instant academic support.
Impact	Solving these challenges will improve learning efficiency, reduce the time spent searching for educational resources, enhance student engagement, support self-paced learning, and improve overall academic performance.
Proposed Solution	
Approach	Develop an AI-powered educational assistant using Google Gemini API integrated with a FastAPI backend. The system processes user queries and generates accurate educational responses, quizzes, summaries, and personalized learning recommendations in real time.
Key Features	• Intelligent Question Answering using Google Gemini. • Simplified Concept Explanations for difficult topics. • AI-Powered Quiz Generation for self-assessment. • Educational Text Summarization. • Personalized Learning Path Recommendations. • Responsive and user-friendly web interface. • Fast and accurate AI-generated educational assistance.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Apple MacBook Air M1 / Intel Core i5 or higher
Memory	RAM specifications	8 GB
Storage	CPU specifications	256 GB SSD or above

Software		
Frameworks	Python frameworks	FlaskAPI
Libraries	Python Libraries	google-generativeai, FastAPI, Uvicorn, python-dotenv, requests
Frontend	User Interface	HTML5, CSS3, JavaScript
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Educational queries and AI-generated responses	User-entered questions, Google Gemini API responses, educational content (JSON/Text format)